Training App — User Audit

Simulated first-time-user walkthrough & capability audit — **all findings resolved**

> **Audited at:** commit `6290d46`, Android debug build on emulator (API 35, GMT)
> **Resolved at:** commit `c042b03`
> **Date:** 28–29 August 2026
> **Method:** new-user task walkthrough on device, cross-checked against the codebase for capabilities that exist in the data layer but are unreachable from the UI
> **Baseline → now:** 179 → 201 automated tests passing, `flutter analyze` clean throughout

> **Status:** every finding in this report has been fixed. The detail below is retained as a record of what was wrong and why it mattered — the reasoning behind each fix is more durable than the fix itself.

This audit simulates a person picking the app up for the first time and trying to do the obvious things: find an exercise, build a plan, train, and look back at what they did. Findings are ordered by how likely they are to block or mislead a real user, not by how hard they are to fix.

Summary of findings

#	Severity	Finding	Area
1	High Fixed	A plan cannot be deleted or renamed	Plans
2	High Fixed	A completed workout cannot be reviewed afterwards	History
3	Medium Fixed	Calendar says “No sessions on this day” before any day is picked	Calendar
4	Medium Fixed	Leaving a session mid-workout silently discards it	Execution
5	Medium Fixed	A library exercise cannot be deleted after creation	Exercise library
6	Medium Fixed	Progression advice always reads in kg, ignoring the lbs setting	Units
7	Low Fixed	Body measurements are cm-only with no imperial option	Units
8	Low Fixed	Imported sessions show a placeholder day (“Any”)	Plans

9	Low Fixed	No search or filter inside a large plan	Plans
10	Fixed	Chart Y-axis labels overlapped near the top	Progress
11	Fixed	Per-exercise history was unreachable from the dashboard	Progress
12	Fixed	History chart axis had no unit label	Progress

The deletion model

Finding #1 was really a symptom. The underlying problem was that deletion was all-or-nothing: the only working delete was *Settings → Delete all data*, so tidying up three junk plans cost you every workout you had ever logged.

The redesign rests on one asymmetry: **plans are cheap, logged workouts are irreplaceable.** A plan can be re-imported from a spreadsheet in a minute. A record of what you lifted last March cannot be recovered by any means. So the model protects history hard and treats plan structure as disposable.

- **Archive is the default, delete is the exception.** Most "make this go away" impulses are really "get it out of my way". Archiving hides a plan, keeps every byte, and is instantly reversible — so it gets an undo snackbar rather than a confirmation dialog. Asking someone to confirm a harmless action only teaches them to dismiss dialogs unread.
- **Deletion states its blast radius in real numbers.** Not "Are you sure?" but *"This plan has 12 logged workouts (148 sets) recorded between 2 March and 20 August."* Nobody can consent to a loss they cannot see.
- **History is separable from the plan that produced it.** Deleting a plan offers to keep the logged workouts: sessions that were actually trained are retained as detached tombstones so history stays readable, while never-trained sessions are dropped. Destroying history is a second, explicit choice — never a side effect of the first.
- **Exercises refuse to vanish silently.** Deleting one first shows where it is used ("appears 3 times in Nine-Block Strength Program") and how much history it carries, removes it from plans, and leaves logged sets intact.
- **Individual workouts can be deleted too.** A mis-logged session no longer requires nuking the plan around it.

Eleven tests in `test/data/plan_deletion_test.dart` pin these semantics — including that an untrained plan leaves nothing behind, that a trained one keeps exactly the sessions with history, and that deleting one plan never touches another's logs.

Detailed findings

1. A plan cannot be deleted or renamed High

What happens: After a few imports the plan list accumulates entries — during this audit the device held *"training_plan_template (second edit)"* and *"neu"* alongside the real programme — with no way to remove or rename any of them.

Why it matters: The only route to a clean list is *Settings → Delete all data*, which also destroys every logged workout. A user tidying up their plan list can lose their entire training history in the process.

Note: `PlanDao.deletePlan()` already exists and is unit-tested, including correct cascade behaviour. This is purely a missing UI entry point, so it is a small change with a large usability payoff.

2. A completed workout cannot be reviewed afterwards High

What happens: Finishing a session writes a full record — every set, rep, weight, RPE and PR flag. None of it is readable afterwards. The only views over history are per-exercise charts and the Progress dashboard.

Why it matters: "What did I actually do on Monday?" is one of the most common questions a trainee asks, and it is the natural way to decide what to load next session. The data is captured but effectively write-only.

Suggested shape: Tapping a completed (green-check) day on the calendar opens that session's log. `SessionLogDao.getLogsForPlanSession()` and `getLoggedSetsForSessionLog()` already return everything needed.

3. Calendar shows a misleading empty state Medium

What happens: On opening the Calendar tab, the panel below reads *"No sessions on this day."* — while today's date visibly carries a scheduled-session dot.

Cause: `_selectedDay` starts as `null`, so the list is empty until the user taps a date, but the message is phrased as though a day had been chosen and found empty.

Fix: Either default the selection to today, or word the no-selection state differently ("Select a day to see its sessions"). The first is likely better — today is almost always what the user wants.

4. Leaving a session mid-workout silently discards it Medium

What happens: Execution mode holds all logged sets in memory and only writes to the database on *Finish session*. Pressing back, or being pulled away by the OS, discards everything with no warning and no recovery.

Why it matters: This was an accepted simplification early on, but the app now actively invites leaving mid-session — the floating rest timer exists precisely so users can switch tabs during a rest. The feature encourages the exact behaviour that loses data.

Fix (cheap): A confirmation prompt on back-navigation when any set is logged. **Fix (better):** Persist sets as they are logged and treat *Finish* as closing the session.

5. A library exercise cannot be edited after creation Medium

Creating a custom exercise captures the full schema, but nothing can be corrected afterwards — a typo in the name, or a missing video link, is permanent. There is also no way to delete a custom exercise, although `ExerciseDao.deleteExercise()` exists. The detail screen is read-only.

6. Progression advice ignores the lbs setting Medium

With the weight unit set to pounds, the auto-progression suggestion on the history screen still reads in kilograms — for example *"try 62.5kg next"* — because `progression.dart`'s `formatWeight()` hard-codes the suffix and does no conversion. Every other weight display converts correctly.

This is a correctness bug, not just cosmetic: a user training in pounds is being handed a number in the wrong unit, with the wrong unit named next to it.

7. Body measurements are centimetre-only Low

Circumference fields are labelled and stored in centimetres regardless of the weight-unit preference. A user who selected pounds has reasonably signalled a preference for imperial units, and will likely expect inches. Bodyweight itself does convert correctly.

8. Imported sessions show a placeholder day Low

Sessions imported without a weekday display as *"Week 1 • Any • Block 1 – Legs"*. "Any" is an internal placeholder surfacing in the UI; it reads as though it were a real choice the user made. Omitting the segment entirely would be clearer.

9. No search inside a large plan Low

The nine-block programme used for testing contains 30+ sessions in a single scrollable list. There is no search, filter, or collapse-by-block. The exercise library has search; plan detail does not, and it is the screen where a long list is most likely.

Fixed during this audit

10–12. Chart and labelling defects Fixed

- **Y-axis label overlap.** The volume chart drew "1K" and "1.1K" on top of each other. Fixed by a shared `niceAxis()` helper using Heckbert "nice numbers" stepping: the axis maximum is now always an exact multiple of the gridline interval, so the top value *is* a tick and no stray label can crowd one. Applied to the history chart as well, replacing its hard-coded version.
- **Dashboard rows were dead ends.** The Progress dashboard listed each exercise's latest top set but could not be tapped, so the trend behind that number was unreachable. Rows now open the exercise's full weight-over-time chart.
- **Unlabelled chart axis.** The history chart's Y axis showed bare numbers with no unit. It now names the active unit in the title.

What held up well

- **Data safety.** Backup/restore round-trips cleanly, including forward-compatibility: a backup taken before body measurements existed still restores. Destructive actions are consistently confirmed.
- **Unit handling.** Weight is stored in kilograms throughout and converted only at display boundaries — the correct architecture, and it holds everywhere except the progression text noted above.
- **Notifications.** The Android 13 permission prompt fires at the right moment, the switch correctly stays off if permission is refused, and scheduled alarms were verified landing at the right wall-clock times via `dumpsys alarm`.
- **Accessibility.** Colour is never the sole carrier of meaning — muscle areas always show their name as text beside the colour — and the palette picker previews real swatches so the choice can be made by eye.
- **Empty states.** Newly added screens explain what to do rather than showing a blank page. The calendar is the one exception, noted above.

How each finding was resolved

#	Resolution	Where
1	Rename / archive / delete, with the impact-aware model above	`plan_actions.dart`, `PlanDao`
2	Workout history list plus a set-by-set detail view, reachable from Progress and from tapping a completed calendar day	`workout_history_screen.dart`
3	Calendar selection defaults to today	`calendar_screen.dart`
4	Back-navigation with unsaved sets now warns, naming how many would be lost	`execution_screen.dart`
5	Custom exercises can be deleted, showing usage first and keeping logged sets	`exercise_detail_screen.dart`
6	Progression text converts and names the active unit; regression-tested	`progression.dart`
7	Circumferences convert to inches when pounds are selected	`units.dart`
8	The internal “Any” placeholder is omitted from display	plan detail, calendar
9	Search over sessions and exercises, shown once a plan exceeds 8 sessions	`plan_detail_screen.dart`

One change beyond the audit

Body-measurement tracking was made **opt-in** rather than always present. Weighing yourself is not a neutral act — plenty of people train deliberately without tracking weight, and for some it is actively unhelpful. The app now asks once, on the way into a first session, and stores the answer as three states rather than two: *not yet asked*, *enabled*, and *declined*. The distinction matters — it means a "no" is remembered as a no and the question is never repeated. It remains available in Settings for anyone who changes their mind, and turning it off hides the feature without deleting anything already recorded.

Training App — local-first training planner. Audited at commit `6290d46`; all findings resolved at `c042b03`. This report is kept as a record of the reasoning, not just the outcome.